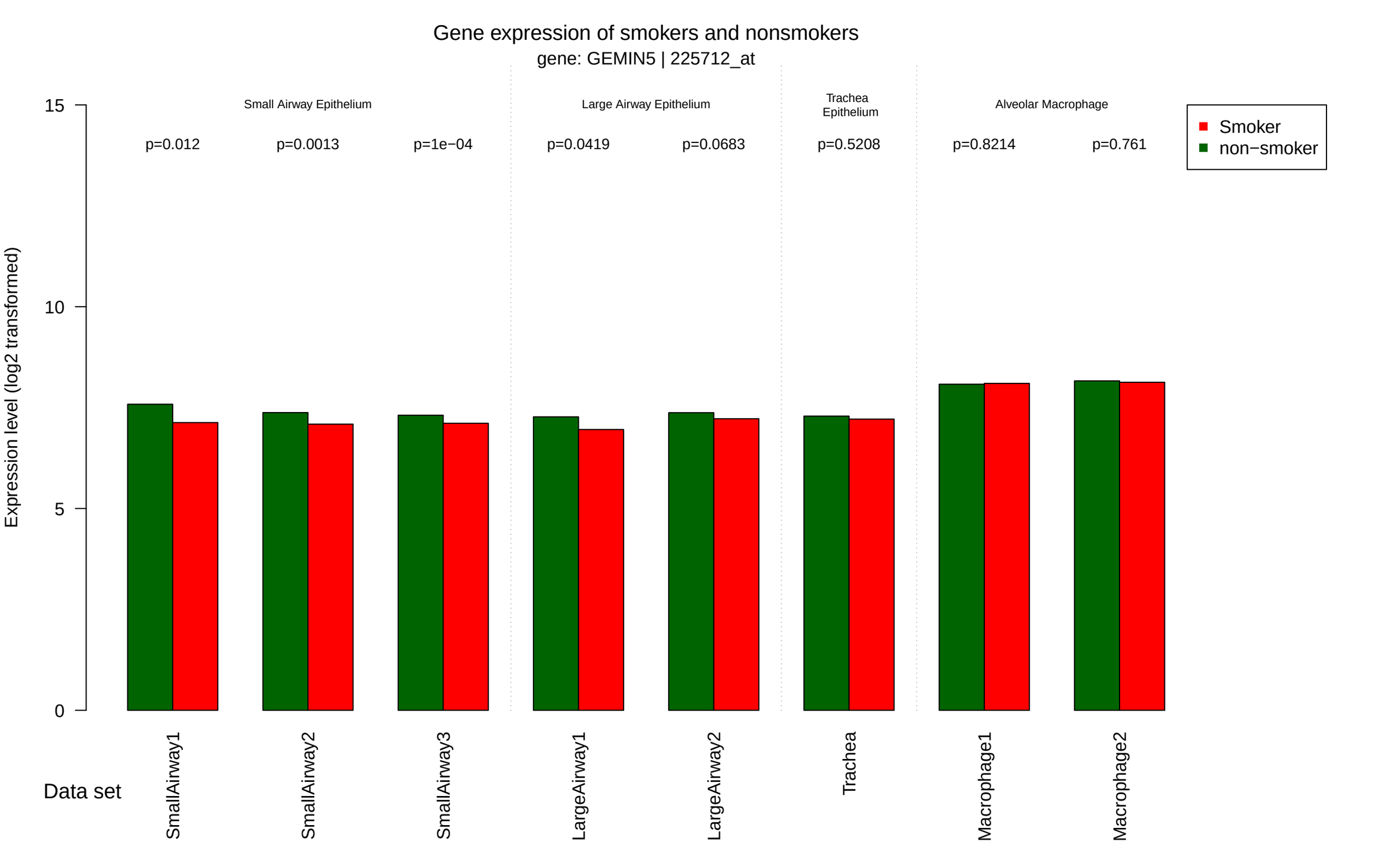
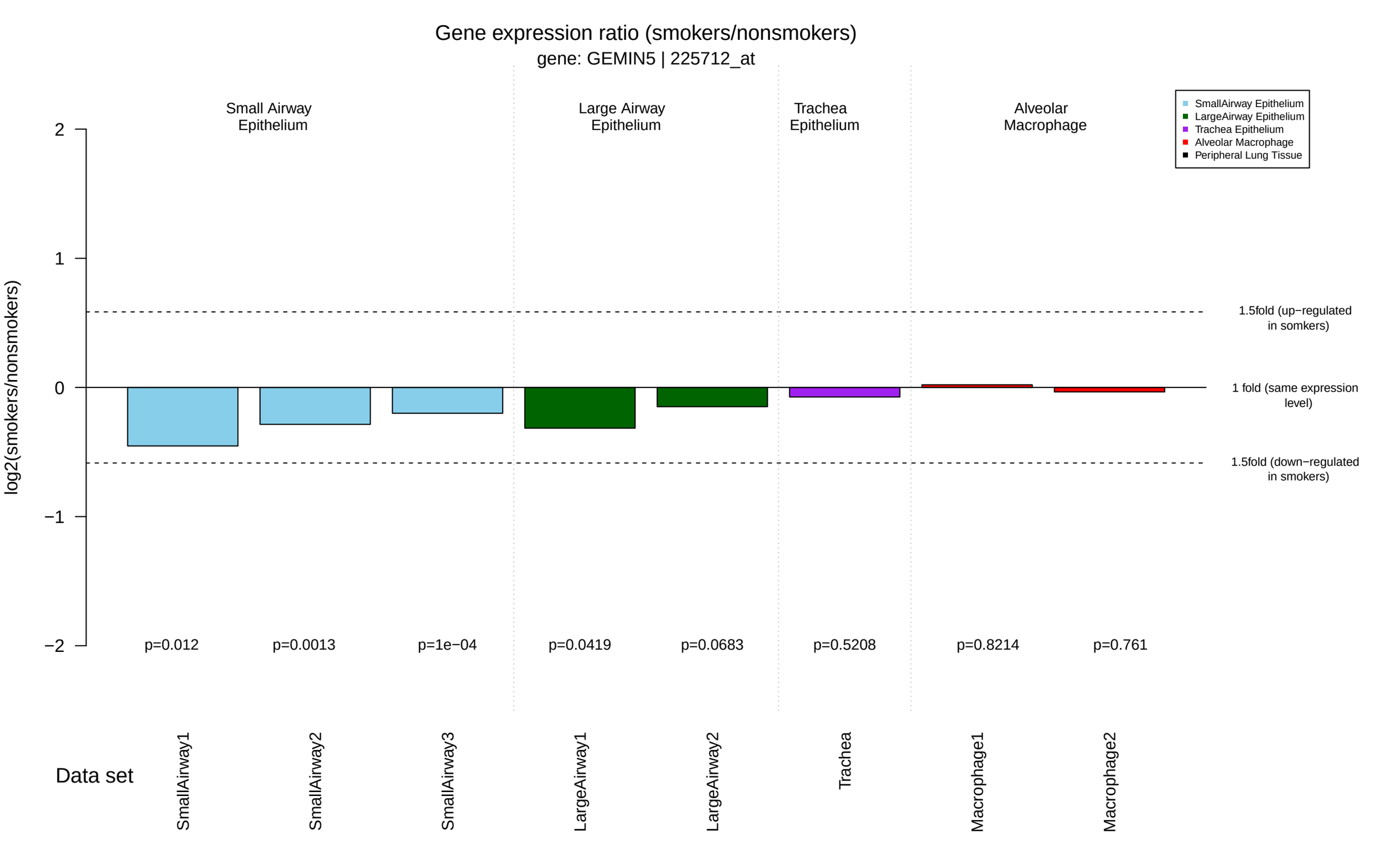
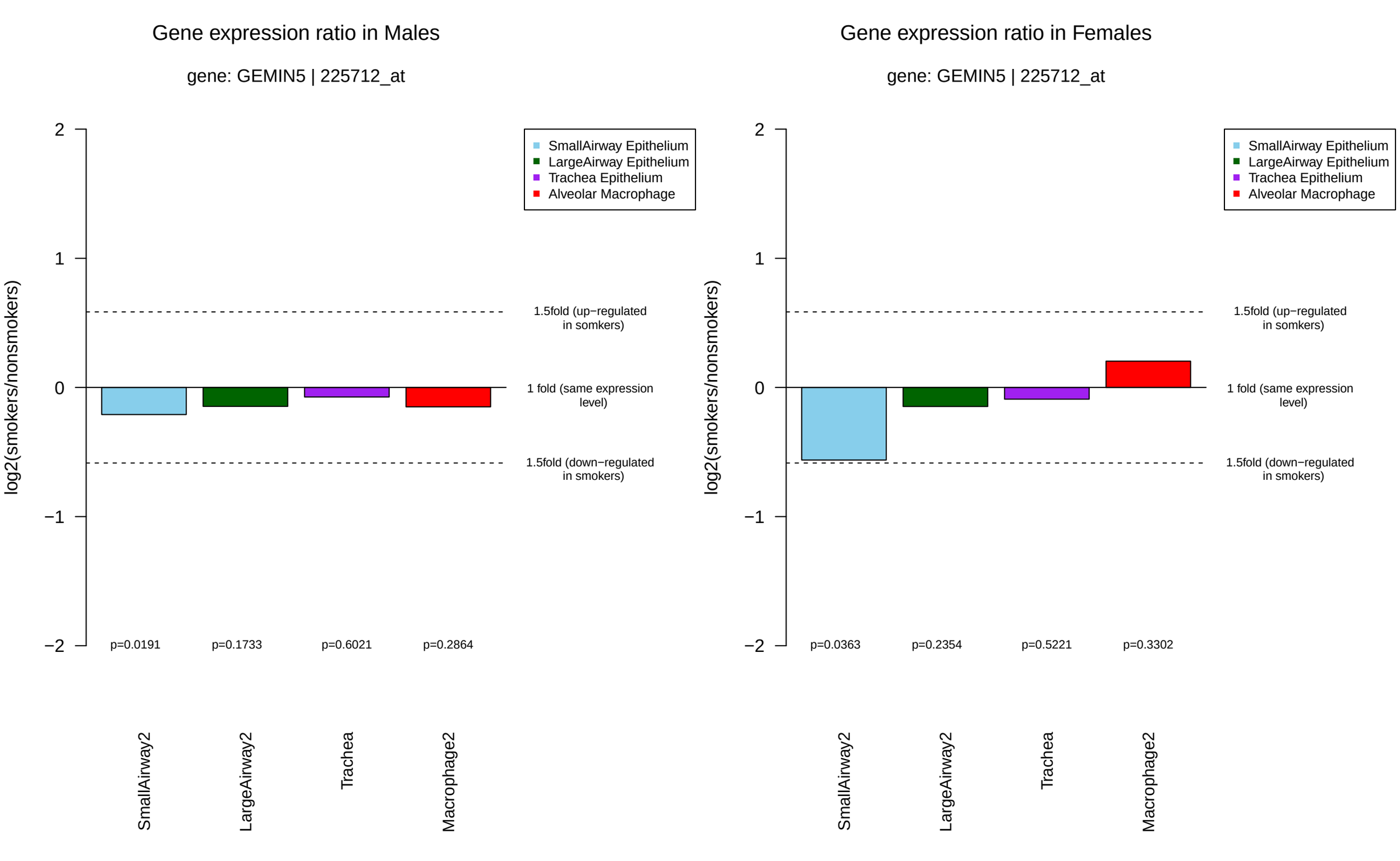
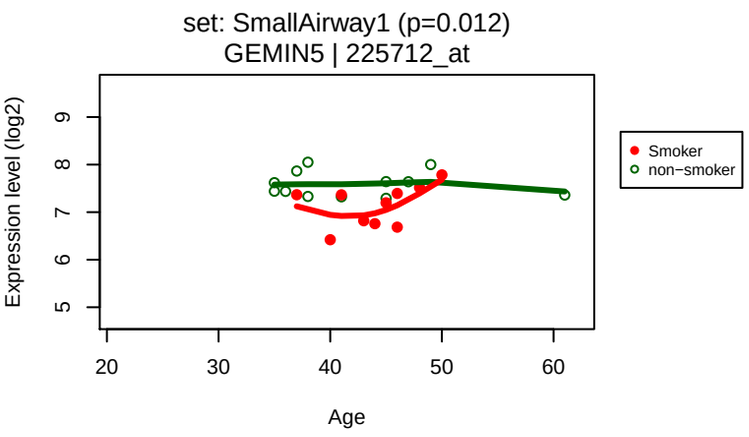
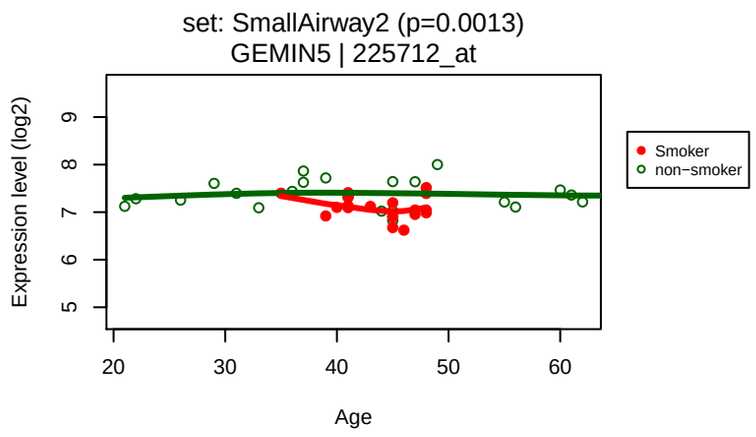
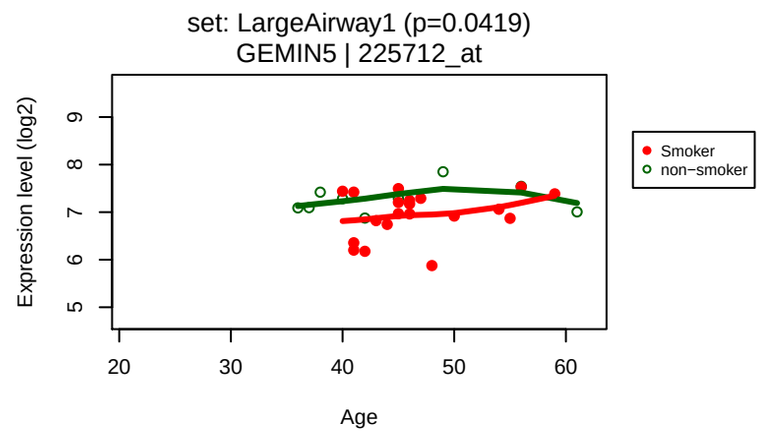

Gene expression of smokers and nonsmokers
gene: GEMIN5 | 225712_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.012
p=0.0013
p=1e−04
p=0.0419
p=0.0683
p=0.5208
p=0.8214
p=0.761
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

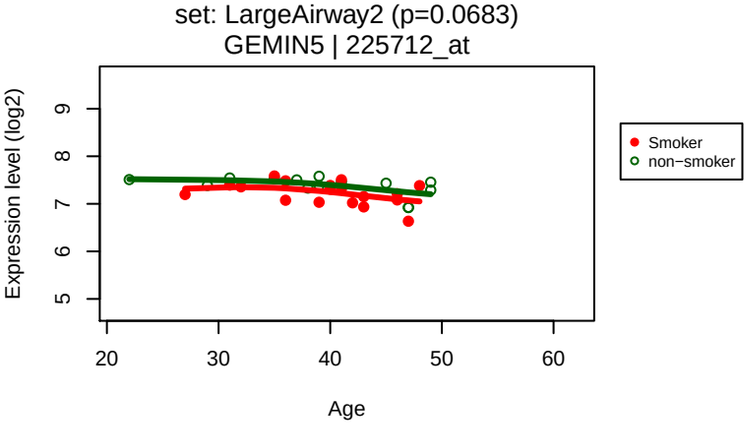

Gene expression ratio (smokers/nonsmokers)
gene: GEMIN5 | 225712_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.012
p=0.0013
p=1e−04
p=0.0419
p=0.0683
p=0.5208
p=0.8214
p=0.761
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

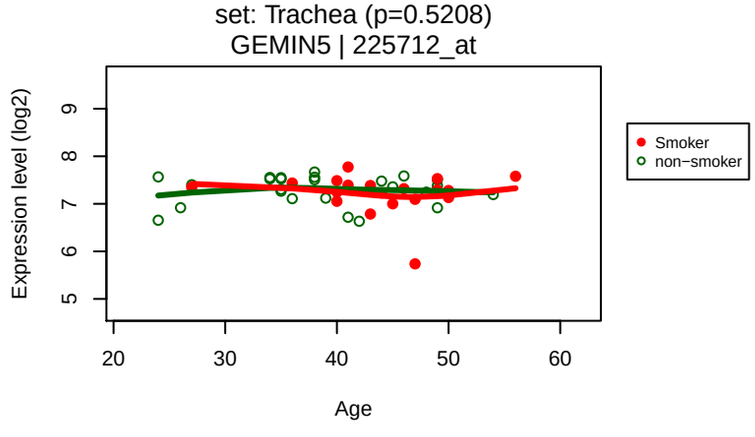

Gene expression ratio in Males
gene: GEMIN5 | 225712_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0191
p=0.1733
p=0.6021
p=0.2864
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GEMIN5 | 225712_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0363
p=0.2354
p=0.5221
p=0.3302
SmallAirway2
LargeAirway2
Trachea
Macrophage2

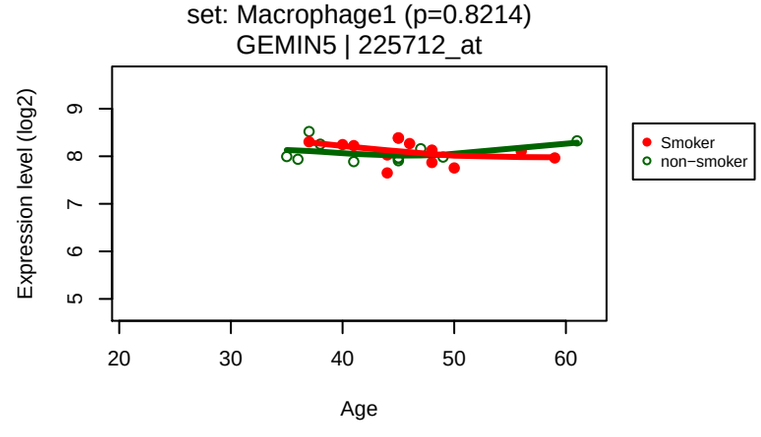

set: SmallAirway1 (p=0.012)
GEMIN5 | 225712_at
Expression level (log2)
Age
Smoker
non−smoker

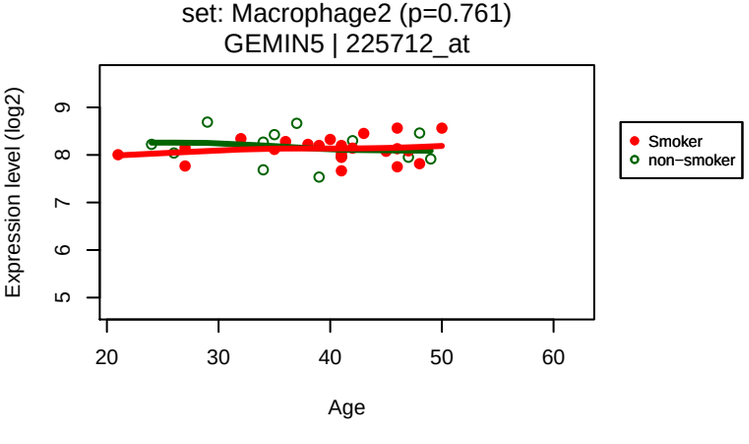

set: SmallAirway2 (p=0.0013)
GEMIN5 | 225712_at
Expression level (log2)
Age
Smoker
non−smoker

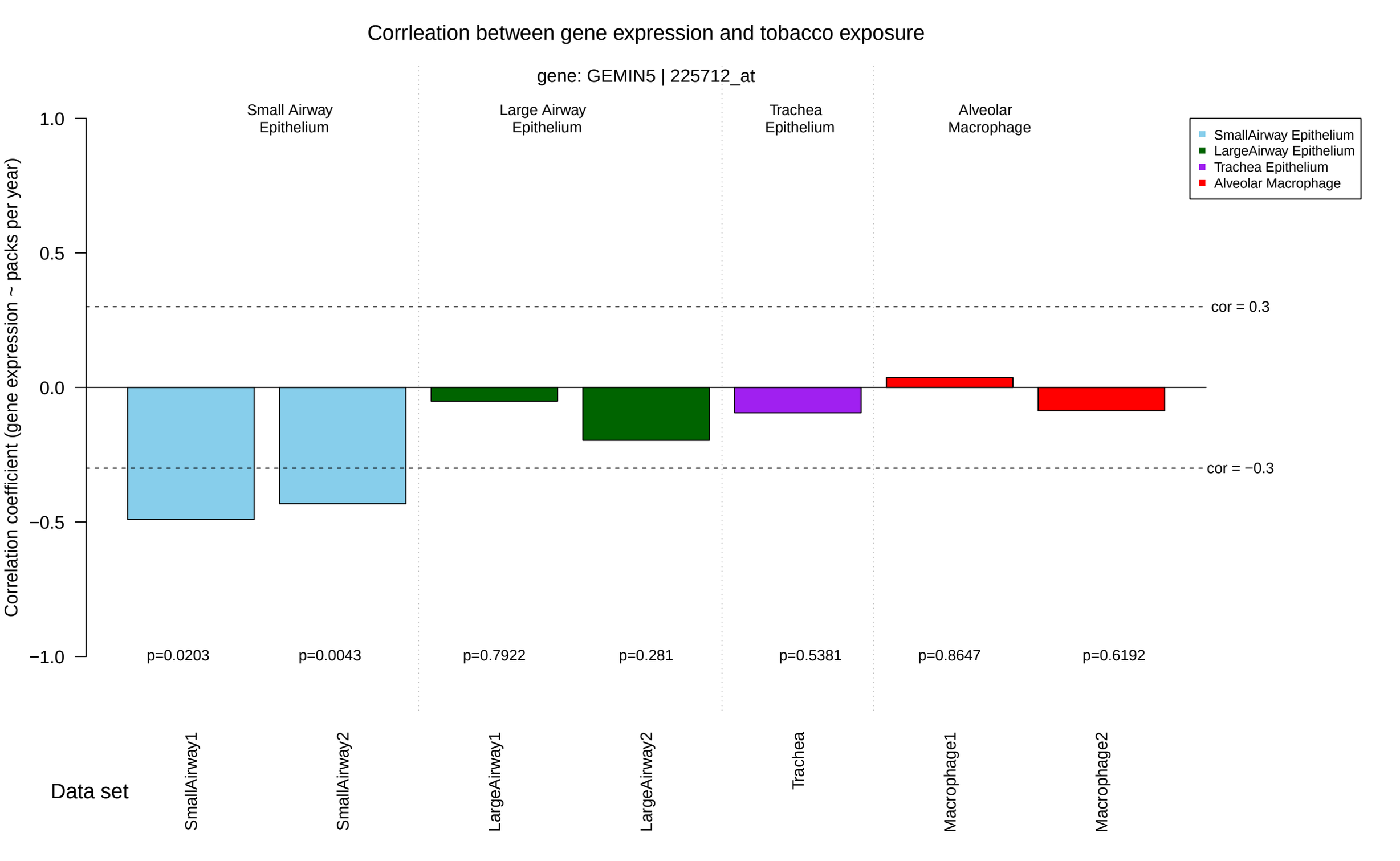

set: LargeAirway1 (p=0.0419)
GEMIN5 | 225712_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0683)
GEMIN5 | 225712_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5208)
GEMIN5 | 225712_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8214)
GEMIN5 | 225712_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.761)
GEMIN5 | 225712_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GEMIN5 | 225712_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0203
p=0.0043
p=0.7922
p=0.281
p=0.5381
p=0.8647
p=0.6192
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2